

Hormuz disruption: 70% — “Current reported”

Bypass assumption: Conservative — EIA spare bypass. Every figure carries {source, url, retrieved, status} in the published dataset. Five pages: exposure · markets · history & structure · the 2026 ladder · method.

12.0 M b/d  
STRANDED — NOWHERE TO GO

14.6 M b/d  
GROSS DISRUPTED — THE HEADLINE NUMBER

2.6/2.6  
BYPASS ABSORBING, M B/D

+77–145%  
BRENT SHOCK BAND — ILLUSTRATIVE

100 days  
STRATEGIC-RESERVE RUNWAY

+33%  
SIEGE PREMIUM, 12–24 MO — PERSISTENT WHILE BARRELS STAY MISSING

10.7 M b/d  
NET GAP AFTER MAX SPR RELEASE — THE TAP BINDS AT 1.3 M B/D

58 Mt/yr  
LNG OFFLINE, ZERO BYPASS — GAS MULTIPLIER ×2.8–×4.6 (ILLUSTR.)

1-in-13-100  
IMPLIED HULL-LOSS ODDS PER TRANSIT, FROM CURRENT QUOTES



| PRODUCER             | EXPORTS | STILL MOVING VS STRANDED                     | STRANDED | EXPOSURE |
|----------------------|---------|----------------------------------------------|----------|----------|
| Saudi Arabia         | 5.5     | <div><div></div><div></div><div></div></div> | -2.3     | PARTIAL  |
| Iraq                 | 3.2     | <div><div></div><div></div><div></div></div> | -2.2     | HIGH     |
| Kuwait               | 1.4     | <div><div></div><div></div><div></div></div> | -1.0     | TOTAL    |
| Qatar                | 1.3     | <div><div></div><div></div><div></div></div> | -0.9     | TOTAL    |
| Iran                 | 1.5     | <div><div></div><div></div><div></div></div> | -0.7     | HIGH     |
| United Arab Emirates | 1.9     | <div><div></div><div></div><div></div></div> | -0.5     | PARTIAL  |
| Bahrain              | 0.2     | <div><div></div><div></div><div></div></div> | -0.1     | HIGH     |

Gulf seaborne exports, M b/d, pre-crisis baselines (EIA country briefs). Per producer: stranded = exports-d – min(own pipeline spare, exports-d). Only Saudi Arabia (Petroline → Yanbu), the UAE (ADCOP → Fujairah) and Iran (Goreh–Jask) own an overland escape.

The number nobody states cleanly: Qatar’s ~77 Mt/yr of LNG — about a fifth of the world’s LNG trade — has zero overland bypass. Petroline can save Saudi barrels; nothing can save Qatari cargoes.

HOW TO READ THIS

Gross vs. net is the whole argument. Headlines quote gross throughput disrupted; the honest damage number subtracts spare overland bypass — capacity that exists **and isn’t already full**. EIA’s conservative spare figure is 2.6 M b/d; post-2025 Petroline-expansion commentary supports up to 5.5, so it ships as a switchable assumption, not a footnote.

The price band is **illustrative by construction** — real markets price expectations, OPEC spare capacity and SPR releases. History’s verdict from 45 years of Hormuz threats (p. 4): spikes mean-revert unless barrels are physically missing. This time they are.

WAR-RISK CONTEXT

Hull war-risk cover for a Gulf transit: **1–7.5%** of vessel value (2026-07) vs **0.25%** pre-crisis — for a \$150M VLCC, **\$1.5M–\$11.3M per voyage**, when cover is offered at all. Full ledger and voyage arithmetic on p. 3.

WHY HORMUZ IS DIFFERENT

The only major oil chokepoint with **no sea alternative**. Suez closed for eight years and the world rerouted around the Cape; here the ships have nowhere else to go (comparison, p. 4).

Importer exposure — who hurts first

An importer's loss = its Hormuz volume × d × (stranded/gross); bypass relief is pro-rata because rerouted barrels are fungible. Days of cover = strategic stocks ÷ lost supply. Sorted shortest first.

| IMPORTER                                   | VIA HORMUZ, M B/D | LOSS, M B/D | % OF ITS DEMAND | STOCKS, M BBL | DAYS OF COVER |
|--------------------------------------------|-------------------|-------------|-----------------|---------------|---------------|
| Other Asia & Pacific <span>ESTIMATE</span> | 6.5               | -3.7        | 42%             | 90            | 24            |
| India                                      | 2.1               | -1.2        | 22%             | 39            | 32            |
| China <span>ESTIMATE</span>                | 5.4               | -3.1        | 19%             | 400           | 129           |
| Rest of world <span>ESTIMATE</span>        | 0.4               | -0.2        | —               | 50            | 217           |
| South Korea                                | 1.7               | -1.0        | 35%             | 240           | 245           |
| Europe (OECD)                              | 2.8               | -1.6        | 12%             | 570           | 354           |
| Japan                                      | 1.6               | -0.9        | 28%             | 470           | 510           |
| United States                              | 0.4               | -0.2        | 1%              | 403           | 999+          |

Volumes: EIA WOTC destinations (2024) — they sum to total throughput, so importer losses sum exactly to stranded barrels (build-enforced). Stocks: IEA / ISPR / DOE; China's are state-opaque — public analyst estimates only. The asymmetry is the story: Asia holds weeks; the OECD holds a year.

RESERVES — THE TAP, NOT THE TANK REPORTED

The stock/flow runway says **100 days** — but coordinated releases max out near **1.3 M b/d** (2022 precedent: 240 M bbl over ~6 months). At today's stranded rate that covers **11%** of the gap, leaving **10.7 M b/d** uncovered. At maximum drawdown the stocks last **923 days** — the tank outlasts the crisis; the tap is the constraint.

MACRO PASS-THROUGH ILLUSTRATIVE

Public rules of thumb (+10% oil ≈ +0.2–0.4 pp CPI, -0.1–0.2 pp GDP over a year) applied to the shock band: consumer prices **+1.5–5.8 pp**; global growth **-0.8–2.9 pp**. A 2-pp global drag ≈ a typical recession's demand loss. Linear rules break down at extreme shocks — treat upper ends as directional. (IMF WEO; Fed staff ranges.)

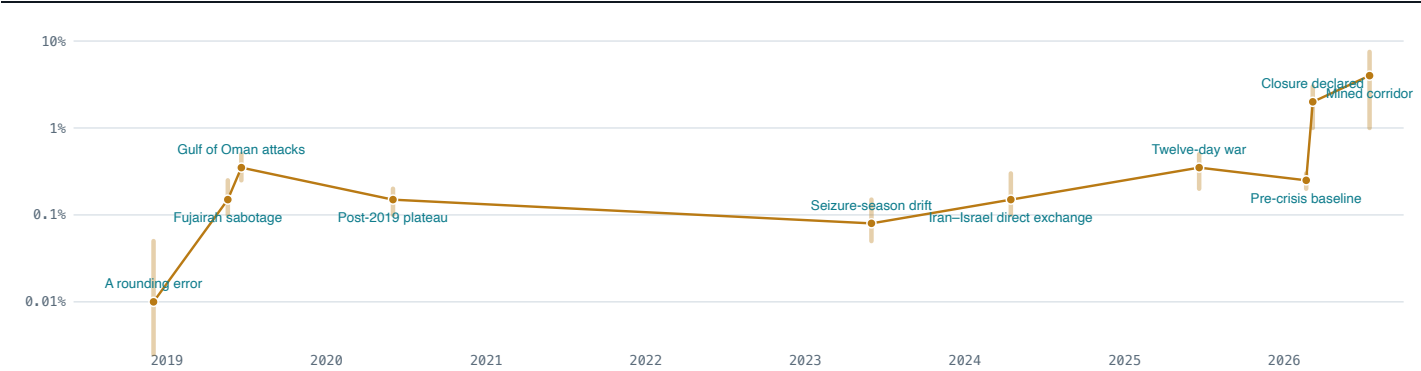
THE LNG SHOCK — NO ESCAPE EXISTS ILLUSTRATIVE

| <b>58 Mt/yr</b>                      | <b>14.5%</b>        | <b>×2.8–×4.6</b>        |
|--------------------------------------|---------------------|-------------------------|
| LNG OFFLINE (7.7 BCF/D)              | OF GLOBAL LNG TRADE | TTF/JKM MULTIPLIER BAND |
| BUYER                                | MT/YR VIA HORMUZ    | LOST AT 70%             |
| Asia (China, India, Korea, Japan...) | 60                  | -42                     |
| Europe                               | 16                  | -11                     |
| Other                                | 7                   | -5                      |

There is no strategic reserve for gas. Europe's storage (~100 bcm, ~25% of annual demand) is a seasonal buffer measured in weeks of winter drawdown; Asian LNG buyers hold days, not months.

The empirical anchor: Russia's 2021–22 pipeline cuts removed roughly the gas-equivalent of 14% of globally traded LNG from Europe's supply — TTF peaked near 10× its pre-crisis level (≈€340/MWh in August 2022 vs ≈€35 in early 2021) before demand destruction bit. VERIFIED

War-risk insurance ledger — % of hull value per Gulf transit (log scale)



JWC listed-area circulars; Lloyd's List, S&P Global and broker commentary. Points inside quoted ranges; bars show the public low-high. From a 0.01% plumbing fee to the number that decides whether a ship sails. Note the ratchet: each crisis's "normal" settles above the last one's.

1-in-13 — 1-in-100

IMPLIED ODDS OF HULL LOSS PER TRANSIT, CURRENT QUOTES

~1-in-10,000

SAME ARITHMETIC, DEC 2018

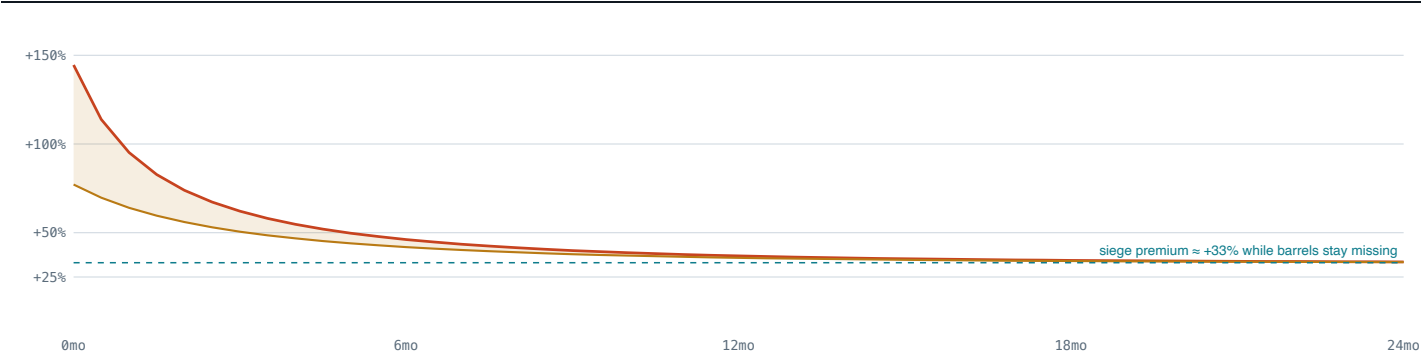
PREMIUM ≈ PROBABILITY × SEVERITY + LOADINGS → UPPER BOUNDS; AT 50% SEVERITY, DOUBLE THE 1-IN-N

Voyage economics — does the transit clear?

| PREMIUM REGIME          | % OF HULL | COST / TRANSIT | \$ / BBL | SHARE OF FREIGHT | VERDICT    |
|-------------------------|-----------|----------------|----------|------------------|------------|
| Feb 2026 “pre-crisis”   | 0.25%     | \$0.3M         | \$0.15   | 4%               | CLEARs     |
| Current — safer vessels | 1%        | \$1.2M         | \$0.60   | 15%              | CLEARs     |
| Current — mid           | 4%        | \$4.8M         | \$2.40   | 60%              | MARGINAL   |
| Current — high-risk     | 7.5%      | \$9.0M         | \$4.50   | 113%             | UNECONOMIC |

\$120M VLCC, 2 M bbl cargo, \$4.0/bbl freight (public broker commentary; crisis chartering has spiked rates 3x+). One-way transit; round trips can double the premium. Above ~100%, the underwriter — not the charterer — decides whether the ship sails.

Dynamic price path — the spike decays; the siege premium stays



$\epsilon(t) = \epsilon_0 + (\epsilon_\infty - \epsilon_0)(1 - e^{-t/\tau})$ ,  $\epsilon_\infty = 0.35$ ,  $\tau = 6$  months ILLUSTRATIVE — assumes the disruption persists; expectations, SPR releases and GDP feedback excluded. Substitution and demand destruction tame the spike, but converge to a persistent premium — not to zero — while barrels stay missing.

Event study — Brent after Hormuz threats (1/5/30 trading days)

| EVENT                                                 | DATE       | +1D    | +5D    | +30D   |
|-------------------------------------------------------|------------|--------|--------|--------|
| Tanker War escalates off Kharg <span>ESTIMATE</span>  | 1984-05-24 | +1.0%  | +2.0%  | -4.0%  |
| Bridgeton mine strike <span>VERIFIED</span>           | 1987-07-24 | -2.8%  | +0.2%  | -10.2% |
| Operation Praying Mantis <span>VERIFIED</span>        | 1988-04-18 | +3.6%  | +1.8%  | -1.5%  |
| Iran closure threat (sanctions) <span>VERIFIED</span> | 2011-12-27 | -1.6%  | +3.7%  | +1.1%  |
| Fujairah sabotage <span>VERIFIED</span>               | 2019-05-12 | +1.0%  | +3.2%  | -11.3% |
| Gulf of Oman attacks <span>VERIFIED</span>            | 2019-06-13 | +2.6%  | +1.9%  | +9.7%  |
| Abqaiq-Khurais strike <span>VERIFIED</span>           | 2019-09-14 | +11.7% | +6.5%  | -4.0%  |
| Soleimani strike <span>VERIFIED</span>                | 2020-01-03 | +3.0%  | -0.7%  | -13.8% |
| Iran strikes Israel directly <span>VERIFIED</span>    | 2024-04-13 | -2.4%  | -5.5%  | -13.0% |
| Israel-Iran twelve-day war <span>VERIFIED</span>      | 2025-06-13 | +7.3%  | +13.5% | +1.7%  |
| 2026 Hormuz crisis begins <span>VERIFIED</span>       | 2026-02-28 | +8.3%  | +34.2% | +70.9% |

EIA Europe Brent spot (RBRTE) via FRED public mirror DCOILBRETEU, daily; windows computed by the fetch pipeline (pre-1987 rows predate the daily series and remain estimates). The pattern: threat spikes mean-revert within weeks — even Abqaiq, the largest single supply disruption in oil-market history, round-tripped in a month — **unless barrels stay missing**. The 2026 row is the counterexample that proves the mechanism: +71% at 30 days and climbing.

Chokepoint anatomy — why Hormuz is different

| CHOKEPOINT         | OIL, M<br>B/D | SEA ALTERNATIVE                  | BYPASS, M<br>B/D | CLOSURE RECORD                                                                                               |
|--------------------|---------------|----------------------------------|------------------|--------------------------------------------------------------------------------------------------------------|
| Strait of Hormuz   | 20.9          | NONE — NO WAY OUT BY SEA         | 2.6              | Never fully closed through 45 years of threats — until 2026. The first actual closure is the current crisis. |
| Strait of Malacca  | 23.7          | Sunda / Lombok straits +1-3d     | 0.4              | Never closed; piracy-era harassment at worst. Rerouting costs days, not barrels.                             |
| Suez Canal + SUMED | 9.2           | Cape of Good Hope +10-14d        | 2.5              | Closed 1956-57 and 1967-1975 (eight years); Ever Given 2021 (six days). The world rerouted and lived.        |
| Bab el-Mandeb      | 8.6           | Cape of Good Hope +10-14d        | —                | 2023-25 Houthi campaign roughly halved transits — traffic diverted, cargoes still arrived.                   |
| Panama Canal       | 0.9           | Cape Horn / Suez routings +5-13d | —                | 2023-24 drought cut transits ~30% — products rerouted at a freight cost.                                     |

EIA WOTC reference volumes. Every other chokepoint has an escape — a longer sea lane, a parallel pipeline, or both. Suez closed for eight years and the world rerouted. Hormuz's first real closure is a different kind of event.

Reopening runway — why d→0 is not day zero

| FRICTION           | PERSISTENCE | ANCHOR                                                                                                                                                                                                                                              |
|--------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Price premium      | 2-6 weeks   | Threat premiums mean-revert within weeks once barrels flow — unless barrels stay missing. Abqaiq 2019: +14.6% one-day spike fully round-tripped in under a month despite 5.7 M b/d offline. <span>VERIFIED</span>                                   |
| War-risk insurance | 4-18 months | Underwriters normalize on renewal cycles and JWC reviews, not headlines — premiums stay multiples of the old baseline for months to years. After the 2019 spiral, Gulf rates held ~10x their April-2019 level well into 2021. <span>ESTIMATE</span> |
| Mined corridor     | 3-9 months  | A swept convoy corridor takes months; certified full clearance takes years. Mines outlast every ceasefire that laid them. 1991 Gulf War: 1,200+ mines; coalition clearance operations ran for over two years. <span>VERIFIED</span>                 |

The road to closure — 2025 → 2026

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-06-13 | <div>Israel strikes Iran — the twelve-day war <span>VERIFIED</span></div> <div>Strikes on nuclear and military targets; Iran answers with missile barrages. Tankers begin GPS-jammed, zig-zag transits.</div> <div>Flows: <b>Transits continue; loadings dip</b> · Price: <b>Brent +~7% day one</b></div>                                                                                                                                   |
| 2025-06-22 | <div>US strikes Fordow; Majlis votes to close the strait <span>VERIFIED</span></div> <div>B-2s hit Fordow, Natanz, Isfahan. Iran's parliament votes (non-binding) to close Hormuz — the decision rests with the SNSC, which doesn't act. The strait stays open.</div> <div>Flows: <b>Self-deterrence: Iran's own exports transit the same water</b> · Price: <b>Brent touches ~\$81 then collapses -7% on ceasefire news (24 Jun)</b></div> |
| 2026-02-28 | <div>The 2026 crisis begins <span>REPORTED</span></div> <div>US and Israeli strikes on Iranian leadership and military targets. Iran's response makes 45 years of closure threats operational for the first time.</div> <div>Flows: <b>Immediate transit hesitancy; insurers pause quotes</b> · Price: <b>+8.3% day one, +34% in a week, +71% in a month (daily spot series) — the first Hormuz spike that did not mean-revert</b></div>    |
| 2026-03-02 | <div>IRGC declares the strait closed <span>REPORTED</span></div> <div>VHF broadcasts warn 'no ship is allowed to pass'; transit permitted only for Iran-approved vessels. Mining of the central corridor and attacks on transiting ships follow.</div> <div>Flows: <b>Tanker transits collapse toward -70%</b> · Price: <b>War-risk quotes move from 0.25% to 1-3% of hull within a week</b></div>                                          |
| 2026-04-11 | <div>US mine-clearing operations begin <span>REPORTED</span></div> <div>Two destroyers transit to 'set conditions' for de-mining; blockade operations against Iranian naval logistics run April-May.</div> <div>Flows: <b>Escorted convoys only; ~150+ vessels damaged or on fire to date (reported)</b> · Price: <b>High-risk hull quotes reach 7.5-10%</b></div>                                                                          |
| 2026-06-15 | <div>Persian Gulf Strait Authority stands up <span>ESTIMATE</span></div> <div>Multinational escort-and-transit authority begins organizing convoys through swept corridors. ~80 mines still reported in the central corridor as of mid-July.</div> <div>Flows: <b>Partial, escorted resumption; ~70% of normal transits still missing</b> · Price: <b>Premiums stay 4x-30x pre-crisis</b></div>                                             |

THE MODEL

|              |                                               |
|--------------|-----------------------------------------------|
| stranded(d)  | = T·d - min(spare_bypass, T·d)                |
| ΔP/P         | ≈ share / ε, ε ∈ [0.08, 0.15], capped         |
| price path   | : ε(t) = ε₀ + (ε∞-ε₀)(1-e <sup>-(t/τ)</sup> ) |
| SPR offset   | = min(max_release_rate, stranded)             |
| producer_i   | : exports_i·d - min(pipeline spare_i, ·)      |
| importer_i   | : imports_i·d · stranded/gross                |
| LNG mult     | : 1 + (lng_lost/global_trade) / ε_gas         |
| implied odds | : P(loss) ≤ premium / severity                |
| voyage       | : share = hull·p / (cargo·freight)            |
| macro        | : (ΔP%/10)·[coeffs], one-year                 |

PROVENANCE LEGEND

**verified** — checked against the named public source at build time · **reported** — live-crisis public reporting · **estimate** — derived/approximate · **illustrative** — model assumption · **brief-baseline** — pending a fresh primary-source check. If a figure is an estimate, it says so.

SOURCES

EIA World Oil Transit Chokepoints (retrieved 2026-07-22); EIA country briefs, STEO and daily Brent · IEA oil security & gas market reports; GIIGNL · ISPRL, DOE, KNOC/METI stock disclosures · JWC listed-area circulars; Lloyd's List, S&P Global and broker commentary · CRS R45281; US Navy histories · 2026-crisis public reporting · IMF WEO / Fed staff rules of thumb · Natural Earth. Per-figure citations with URLs and retrieval dates ship in the dataset and render in the site's Methodology panel.

INTENDED USE

Decision-support for shipping, energy and policy analysis from 100% public data. No real-time vessel positions, no navigational data, nothing that could serve as a targeting aid. Incident positions are approximate; vessel-behavior content is pattern-level.

Interactive version, all scenarios: **chokepoint.analyticadss.com**  
Scenario deep link: <http://localhost:5199/?intro=skip&brief=1#v=map&d=70&b=eia&y=2026&e=11>  
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